UNIVERSITY OF CALIFORNIA SAN DIEGO

Generative AI for Music and Audio

A dissertation submitted in partial satisfaction of the
requirements for the degree Doctor of Philosophy

in

Computer Science

by

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The Dissertation of Hao-Wen Dong is approved, and it is acceptable in quality and form for publication on microfilm and electronically.

University of California San Diego

2024

DEDICATION

To my wife Eltha, my parents Nancy and David, and my pooh bear, without whom nothing would have been possible.

EPIGRAPH

Without music, life would be a mistake.

—Friedrich Nietzsche

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- *Chapter 2, in full, is a reprint of the material as it appears in "MusPy: A Toolkit for Symbolic Music Generation" by Hao-Wen Dong, Ke Chen, Julian McAuley and Taylor Berg-Kirkpatrick, which was published in the Proceedings of the International Society for Music Information Retrieval Conference (ISMIR) in 2020. The dissertation author was the primary investigator and author of this paper.*
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- *Chapter 4, in full, is a reprint of the material as it appears in "Towards Automatic Instrumentation by Learning to Separate Parts in Symbolic Multitrack Music" by Hao-Wen Dong, Chris Donahue, Taylor Berg-Kirkpatrick and Julian McAuley, which was published in the Proceedings of the International Society for Music Information Retrieval Conference (ISMIR) in 2021. The dissertation author was the primary investigator and author of this paper.*
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Hao-Wen Dong, Xiaoyu Liu, Jordi Pons, Gautam Bhattacharya, Santiago Pascual, Joan Serrà, Taylor Berg-Kirkpatrick, and Julian McAuley, “CLIPSonic: Text-to-Audio Synthesis with Unlabeled Videos and Pretrained Language-Vision Models,” *IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, 2023.

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Hao-Wen Dong, Cong Zhou, Taylor Berg-Kirkpatrick, and Julian McAuley, “Deep Performer: Scoreto-Audio Music Performance Synthesis,” *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022.

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Hao-Wen Dong[*], Wen-Yi Hsiao[*], Li-Chia Yang, and Yi-Hsuan Yang, “MuseGAN: Multi-Track Sequential Generative Adversarial Networks for Symbolic Music Generation and Accompaniment,” *AAAI Conference on Artificial Intelligence (AAAI)*, 2018. ([*]equal contribution)

Hao-Wen Dong[*], Wen-Yi Hsiao[*], Li-Chia Yang, and Yi-Hsuan Yang, “MuseGAN: Demonstration of a Convolutional GAN Based Model for Generating Multi-track Piano-rolls,” *ISMIR Late-Breaking Demos*, 2017. ([*]equal contribution)

ABSTRACT OF THE DISSERTATION

Generative AI for Music and Audio

by

Hao-Wen Dong

Doctor of Philosophy in Computer Science

University of California San Diego, 2024

Professor Taylor Berg-Kirkpatrick, Co-Chair
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Generative AI has been transforming the way we interact with technology and consume content. In the next decade, AI technology will reshape how we create audio content in various media, including music, theater, films, games, podcasts, and short videos. In this dissertation, I introduce the three main directions of my research centered around *generative AI for music and audio*: 1) multitrack music generation, 2) assistive music creation tools, and 3) multimodal learning for audio and music. Through my research, I aim to answer the following two fundamental questions: 1) *How can AI help professionals or amateurs create music and audio content?* 2) *Can AI learn to create music in a way similar to how humans learn music?* My long-term goal is to lower the barrier of entry for music composition and democratize audio content creation.

Chapter 1

Introduction

AI is the study of how to make computers do things
at which, at the moment, people are better.
—Elaine Rich and Kevin Knight

Generative AI has been transforming the way we interact with technology and consume content. The recent success of large language model-based chatbots (e.g., OpenAI's ChatGPT[1] and Google's Gemini[2]), AI assistants (e.g., GitHub and Microsoft Copilot) and text-to-image generation systems (e.g., Adobe Firefly,[3] Midjourney[4] and Stable Diffusion[5]) showcases how AI-powered technology can be integrated into professional workflows and boost human productivity. In the next decade, generative AI technology will also reshape how we create audio content in the $2.3 trillion global entertainment industry, including the music, film, TV, podcast and gaming sectors. Take AI-powered music creation for example: On one hand, we have witnessed major progress in automatic music composition (Briot et al., 2017; Huang et al., 2020), which has long been considered as a grand challenge of AI. On the other hand, our expectations of *AI Music* today has expanded to cover the whole music creation process—from composition, arrangement, sound production, recording to mixing (De Man et al., 2019). With a growing momentum in both academia and industry, AI-powered audio creation has been gaining attention in the broader AI community.

[1] https://chat.openai.com/
[2] https://gemini.google.com/
[3] https://firefly.adobe.com/
[4] https://www.midjourney.com/
[5] https://stability.ai/stable-image

Figure 1.1. Looking back to the past, how music and technology interacts has always been a two-way process. (Left) the violin-making industry grows with the classical music, and together create the golden age of classical music. (Right) the invention and development of synthesizers and drum machines helped popularize electronic music. Image sources (from left to right): 1) Mozart83, Public domain, via Wikimedia Commons, 2) Hildegard Dodel, Public domain, via Wikimedia Commons, 3) yan, CC BY-SA 4.0, via Wikimedia Commons, and 4) taken at Hamamatsu Museum of Musical Instruments, August 2019.

My research springs from two fundamental questions: 1) *How can AI help professionals or amateurs create music and audio content?* 2) *Can AI learn to create music in a way similar to how humans learn music?* From a musical perspective, technology has always been a driving factor of music evolution. For example, the study of acoustics and musical instrument making fostered the development of classical music; the invention of synthesizers and drum machines helped popularize electronic music. I am thus interested in exploring how the latest AI technology can empower artists to create novel contents. From a technical perspective, music possesses a unique complexity in that music follows rules and patterns while being creative and expressive at the same time. I am thus fascinated about the idea of building intelligent systems that can learn, create and play music like humans do. I envision the future development of AI Music to be a two-way process—*new technology creates new music; new music inspires new technology*.

Motivated by this belief, I study a wide range of topics centered around *Generative AI for Music and Audio*, including multitrack music generation (Dong et al., 2018a; Dong et al., 2017; Dong and Yang, 2018; Dong et al., 2023a; Xu et al., 2023; Liu et al., 2018a), automatic instrumentation (Dong et al., 2021), automatic arrangement (Dong et al., 2018a; Liu et al., 2018a), automatic harmonization (Yeh et al., 2021), music performance synthesis (Dong et al.,

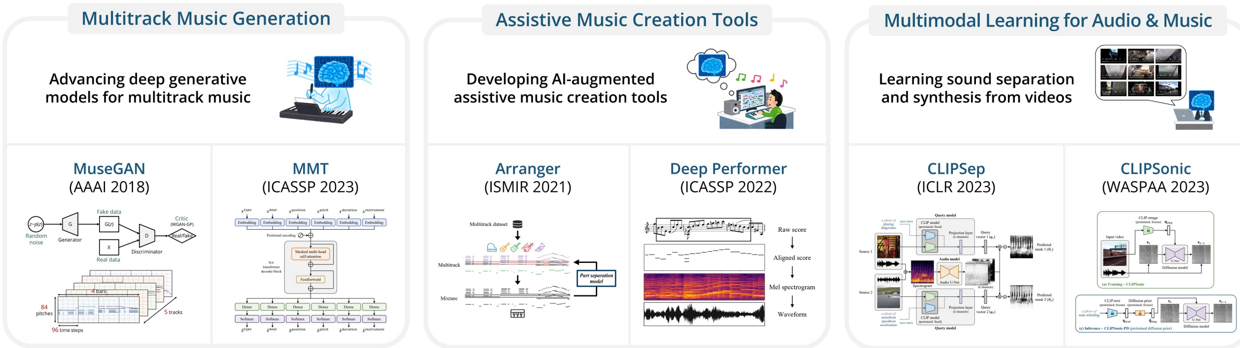

Figure 1.2. An overview of the three main directions of my research.

2022), text-queried sound separation (Dong et al., 2023d), text-to-audio synthesis (Dong et al., 2023c; Dong et al., 2023b) and symbolic music processing software (Dong et al., 2018b; Dong et al., 2020).

My research can be categorized into three main directions, as shown in Figure 1.2: 1) **multitrack music generation**—*advancing deep generative models for multitrack music*, 2) **assistive music creation tools**—*developing AI tools that can help musicians and amateurs create music*, and 3) **multimodal learning for audio and music**—*learning sound separation and synthesis from noisy videos*.

1.1 Multitrack Music Generation

Researchers have been working on automatic music composition for decades, and it has long been viewed as a grand challenge of AI. I started this thread of research on multitrack music generation in 2017. Back then, prior work on deep learning-based music generation had focused on generating melodies, lead sheets (i.e., melodies and chords) or four-part chorales Briot et al., 2017. However, modern pop music often consists of multiple instruments or tracks. To make deep learning technology applicable in modern music production workflow, it is important to modernize deep learning models for multitrack music generation.

Witnessing the lack of infrastructure for symbolic music generation when conducting these research projects, I developed libraries for processing symbolic music to consolidate the infrastructure of music generation research (Dong et al., 2018b; Dong et al., 2020). The Python toolkits I developed to process symbolic music for machine learning applications have been widely used in the field. With these libraries, researchers can easily download commonly used datasets programmatically and be liberated from reimplementing tedious data processing routines. The toolkits also allowed me to conduct *the first large-scale experiment that measures the cross-dataset generalizability of deep neural networks for music* (Dong et al., 2020). This work will be presented in Chapter 2.

Moreover, I study efficient music generation model with an eye to enabling real time improvisation or near real time creative applications. In Dong et al., 2023a, I proposed the Multitrack Music Transformer (MMT) model that achieves comparable performance with state-of-the-art systems, landing in between two recently proposed models in a subjective listening test, while achieving substantial speedups and memory reductions over both. Further, I presented *the first systematic analysis of musical self-attention*, where I showed that the trained model learns a relative attention in certain aspects of music. With the great success brought by large-scale generative pretraining in natural language processing, my ongoing work aims to scale this method up using the largest ever symbolic music dataset containing more than one million scores, which I collected and compiled from the MuseScore forum. This work will be presented in Chapter 3.

1.2 Assistive Music Creation Tools

Music creation today is still largely limited to professional musicians for it requires a certain level of knowledge in music theory, music notation and music production tools. Apart from generating new music content from scratch, another line of my research focuses on developing AI-augmented tools to assist amateurs to create and perform music. My long-term goal along this research direction is to lower the entry barrier of music composition and make music creation accessible to everyone.

For example, in (Dong et al., 2021), I developed *the first deep learning model for automatic instrumentation*. Instrumentation refers to the process where a musician arranges a solo piece for a certain ensemble such as a string quartet or a rock band. This can be challenging for amateur composers as it requires domain knowledge of each target instrument. In this work, I proposed a new machine learning model that can produce convincing instrumentation for a solo piece by framing this problem as a sequential multi-class classification problem. Such an automatic instrumentation system can suggest potential instrumentation for amateur composers, especially useful when arranging for an unfamiliar ensemble. Further, the proposed model can empower a musician to play multiple instruments on a single keyboard at the same time. This work will be presented in Chapter 4.

Another example is my work on music performance synthesis (Dong et al., 2022). While synthesizers play a critical role and are intensively used in modern music production, existing synthesizers either requires an input with expressive timing or allows only monophonic inputs. In light of the similarities between text-to-speech (TTS) and score-to-audio synthesis, I showed in this work that we can adapt a state-of-the-art TTS model for music performance synthesis. Moreover, I proposed a novel mechanism to enable polyphonic music synthesis. This work represents *the first deep learning based polyphonic synthesizer that can synthesize a score into a natural, expressive performance*. This work will be presented in Chapter 5.

1.3 Multimodal Learning for Audio and Music

The third line of my research focuses on multimodal learning for audio and music. Sound is an integral part of movies, dramas, documentaries, podcasts, games, short videos and audiobooks. In these media, audio and music production tools need to interact with inputs from other modalities such as text and images, and thus multimodal models are critical in enabling controllable creation tools for music and audio in these applications.

Along this direction, I have worked on text-queried sound separation (Dong et al., 2023d) and text-to-audio synthesis (Dong et al., 2023c; Dong et al., 2023b). Unlike existing work that relies on a large amount of paired audio-text data, I explore a new direction of approaching bimodal learning for text and audio through leveraging the visual modality as a bridge. The key idea behind my study is to combine the naturally-occurring audio-visual correspondence in videos and the multimodal representation learned by contrastive language-vision pretraining (CLIP). Based on this idea, I developed *the first text-queried sound separation model that can be trained without any text-audio pairs* (Dong et al., 2023d). Text-queried sound separation aims to separate a specific sound out from a mixture of sounds given a text query, which has many downstream applications in audio post-production